TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

AI.104

Computer Vision

COURSE OVERVIEW

How machines see: image formation, filtering, feature detection, and introductory object recognition.

1

SESSION

60 MINUTES

ORIENTATION: THE MACHINE EYE

KEY TOPICS

- Image formation and the camera model
- Pixels, color spaces, and digital images
- The computer vision pipeline
- Vision in robotics and AI

TEACHING FOCUS

Understand how images become machine-readable data.

KEY TAKEAWAY Seeing begins with light, lenses, and sensors.

2

SESSION

60 MINUTES

IMAGE PROCESSING & FILTERING

KEY TOPICS

- Point operations: brightness, contrast
- Convolution and spatial filtering
- Smoothing: Gaussian and median
- Edge detection: Sobel, Canny

TEACHING FOCUS

Enhance and extract structure from images.

▲ **KEY TAKEAWAY** Filters reveal what the eye may miss.

COMPUTER VISION

3

SESSION

60 MINUTES

FEATURES & KEYPOINTS

KEY TOPICS

- Corners: Harris and Shi-Tomasi
- SIFT and ORB descriptors
- Feature matching and tracking
- Template matching

TEACHING FOCUS

Identify stable points across views.

KEY TAKEAWAY Keypoints anchor vision to the world.

4

SESSION

60 MINUTES

IMAGE SEGMENTATION

KEY TOPICS

- Thresholding and Otsu
- Region growing and watershed
- Contour detection
- Semantic vs. instance segmentation

TEACHING FOCUS

Partition an image into meaningful regions.

KEY TAKEAWAY Segmentation separates object from background.

5

SESSION

60 MINUTES

OBJECT RECOGNITION BASICS

KEY TOPICS

- Sliding-window detection
- Histogram of oriented gradients
- Intro to deep vision
- Face and object recognition

TEACHING FOCUS

Name and locate things in images.

KEY TAKEAWAY Recognition answers 'what is where?'.

COMPUTER VISION

6

SESSION

60 MINUTES

VISION FOR ROBOTS & REVIEW

KEY TOPICS

- Visual servoing and motion
- Camera calibration
- Course recap and assessment
- Toward machine vision (AI.307)

TEACHING FOCUS

Connect vision to robot action.

▲ **KEY TAKEAWAY** Robots act on what they see.